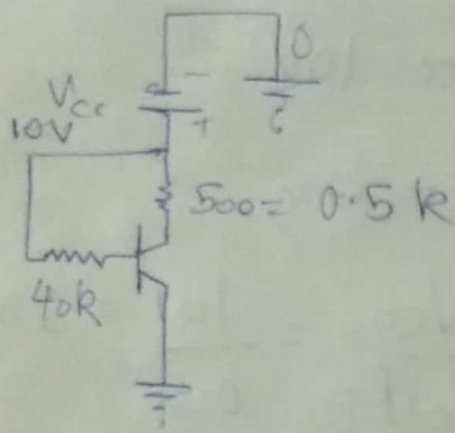
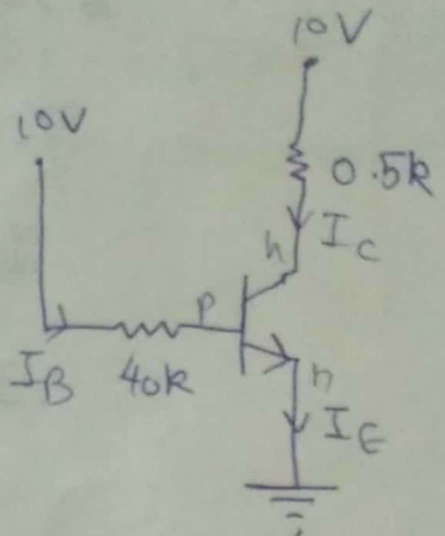
# Assignment of BJT

1.)

$\beta_{dc} = 5$




$\longleftrightarrow$




i/p KVL –

$10 - 40 I_B - 0.7 = 0$

$\Rightarrow \quad I_B = \frac{9.3}{40} \text{ mA} = 0.2325 \text{ mA}$

$I_c = \beta I_B = 50 \times 0.2325 = \cancel{6.975 mA}$

$11.625 mA$

$I_E = I_B + I_C = \cancel{7.2075 mA}$

$11.8575 mA$

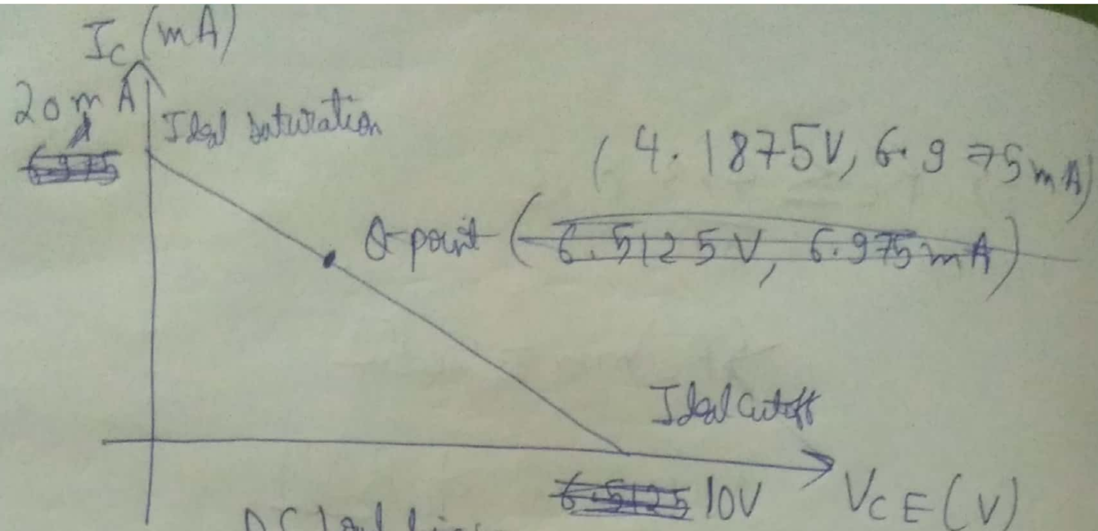
o/p loop KVL —

$10 - 0.5 \times \overset{11.625}{\cancel{6.975}} - V_{CE} = 0$

$\because V_{CB} = V_{CE} - V_{BE} = \overset{3.48}{\cancel{4.4875}} V > 0 \Rightarrow \boxed{\cancel{V_{CE} = 6.5125 V}} \quad \boxed{V_{CE} = 4.1875 V}$

Q-point is ~~(6.5125 V, 6.975 mA)~~

(4.1875 V, 6.975 mA)




DC Load line:

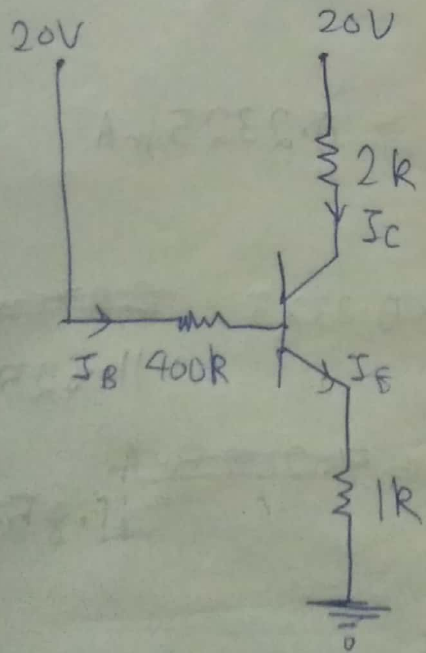
For ideal cut-off,

$$V_{CE}(off) = V_{CC} = 10V$$

For ideal saturation,

$$I_{C(sat)} = \frac{V_{CC}}{R_C+R_E} = \frac{10}{0.5} = 20mA$$

2.)




Sol$^n$: given, $\beta_{dc} = 50$

and $I_C \neq I_E$

i/p loop KVL-

$$20 - 400I_B - 0.7 - (\beta+1)I_B = 0$$

$$\Rightarrow 19.3 = 451 I_B$$

$$\Rightarrow \boxed{I_B = 0.0428mA}$$

o/p loop KVL-

$$20 - 2\times50\times0.0428 - V_{CE} - 51\times0.0428 = 0$$

$$\Rightarrow \boxed{V_{CE} = 13.5372V}$$

Scanned by CamScanner

$\because V_{CB} = V_{CE} - V_{BE}$

$= 12.8372V > 0$

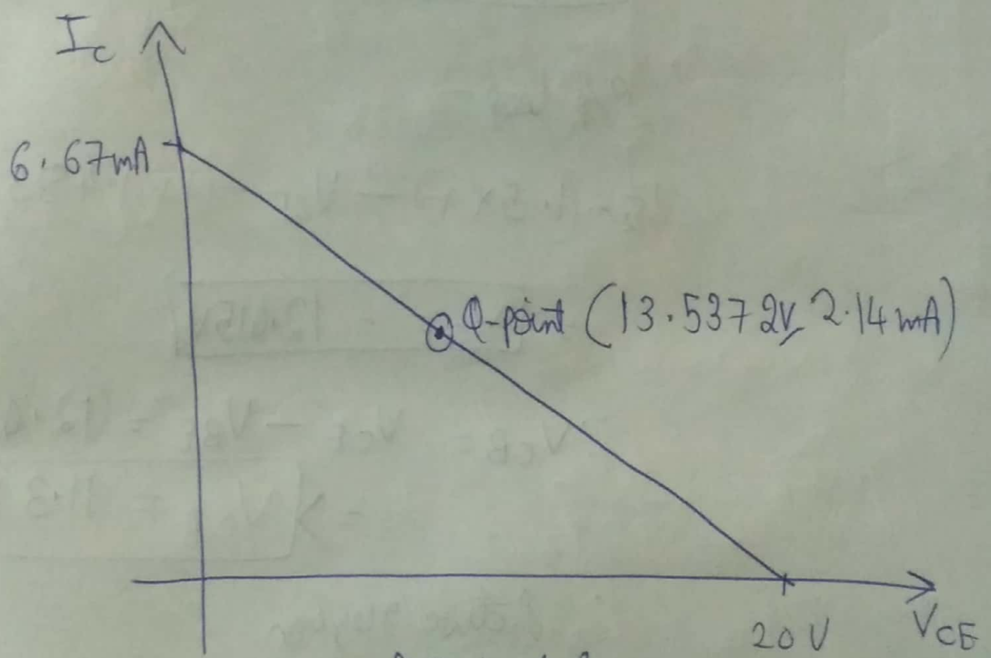
$\therefore$ Active region

$I_C = 50 \times I_B = 2.14mA$

For cutoff, $V_{CE(off)} = V_{CC} = 20V$

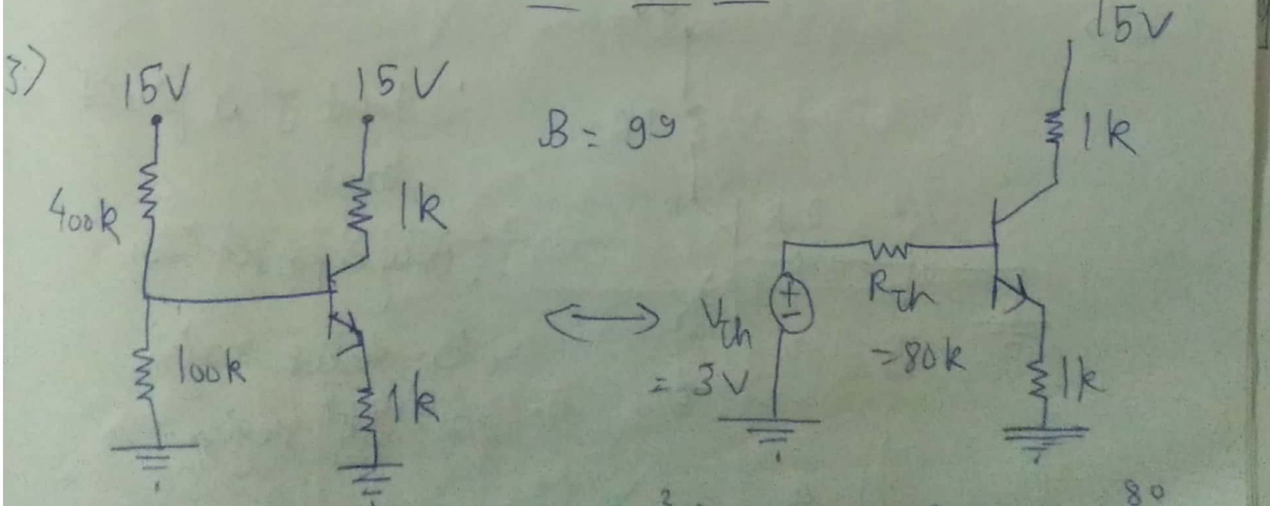
For sat$^n$,

$$I_C(sat) = \frac{V_{CC}}{2+1} = \frac{20}{3} = 6.67mA$$




DC Load line

3)




$V_{th} = \frac{15}{5} \times 1 = 3V$, $R_{th} = \frac{100 \times 400}{500} = 80k$

i/p loop —

$$3 - 80I_B - 0.7 - 100I_B = 0$$

$$\Rightarrow I_B = \frac{2.7}{180}$$

$$\Rightarrow \boxed{I_B = 0.015mA}$$

$$\boxed{I_C = 1.485mA}$$

$$\boxed{I_E = 1.500mA}$$

o/p loop —

$$15 - (1.5 \times 1) - V_{CE} - (1 \times 1.485) = 0$$

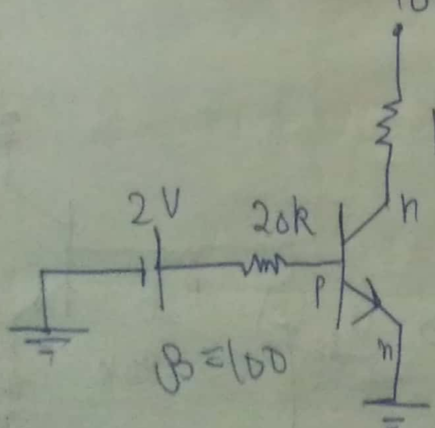
$$\Rightarrow \boxed{V_{CE} = 12.015V}$$

$$\because V_{CB} = V_{CE} - V_{BE} = (12.015 - 0.7)V$$

$$\Rightarrow \boxed{V_{CB} = 11.315V}$$

$\therefore$ Active region

4.)




$\because$ Input $J^n$ is forward bias

$\therefore$ Only 2 possibilities

① Active region

② Satn region.

Let us assum active region

Scanned by CamScanner

i/p loop.

$$2V - 20 \times I_B - 0.7 = 0$$

$$\Rightarrow \boxed{I_B = 0.065 mA}$$

$$\boxed{I_C = 100 \times I_B = 6.5 mA}$$

By KCL, $\boxed{I_E = 6.565 mA}$

O/p loop-

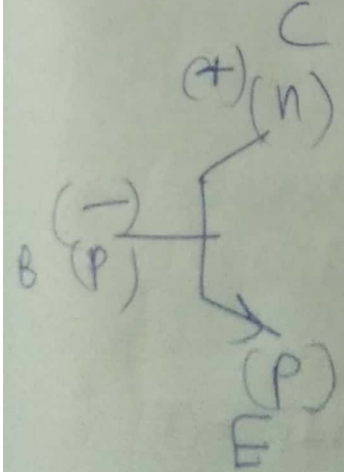
$$10 - 1 \times 6.5 - V_{CE} = 0$$

$$\Rightarrow \boxed{V_{CE} = 3.5V}$$

Now, $V_{CB} = V_{CE} - V_{BE}$

$$= (3.5 - 0.7)V$$

$$= 2.8V$$




$\therefore V_{CB} > 0V$

Clearly, O/p $J^n$ is reverse bias

$\therefore$ Our assumption is correct

Hence, Active Region.

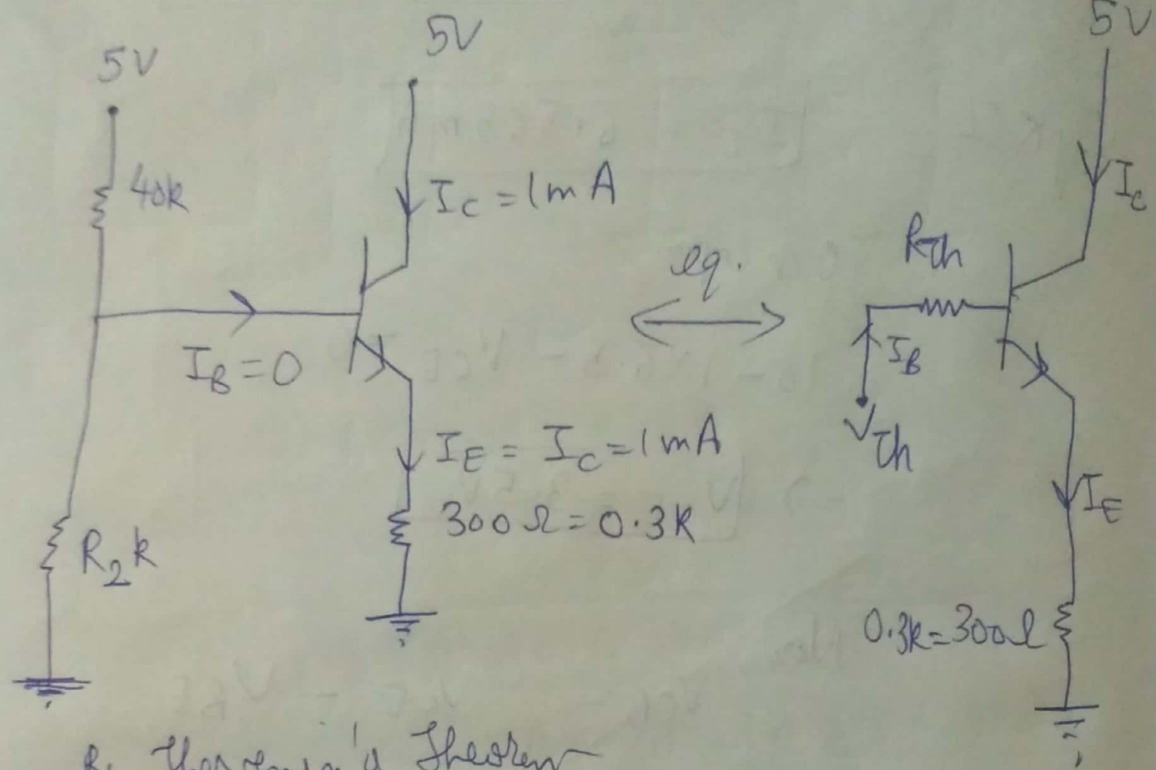
Scanned by CamScanner

5.) Given,

$I_c = 1mA$

$\beta_{dc} = \infty \Rightarrow \boxed{I_B = 0} \Rightarrow \boxed{I_C = I_E}$

$V_{BE} = 0.7V$




By Thevenin's Theorem

$$V_{th} = \frac{5R_2}{40+R_2}$$

$$R_{th} = \frac{40R_2}{40+R_2}$$

$\therefore$ i/p KVL-

$$V_{th} - 0 \times R_{th} - 0.7 - 0.3 \times I_E = 0$$

$$\Rightarrow V_{th} = 0.3 \times 1 + 0.7$$

$$\Rightarrow \frac{5R_2}{40+R_2} = 1 \Rightarrow 4R_2 = 40$$

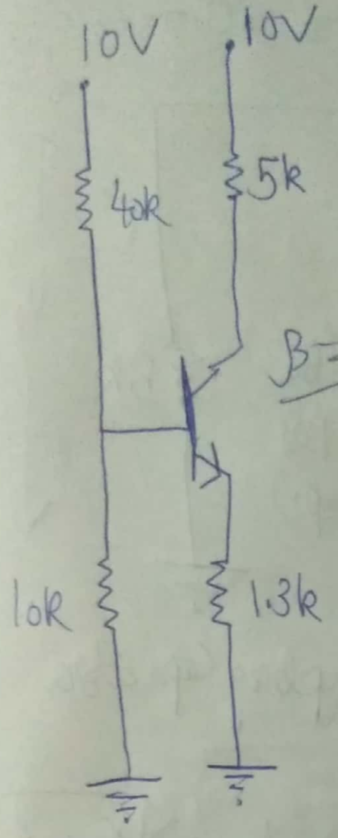
$$\Rightarrow \boxed{R_2 = 10k\Omega}$$

6.7

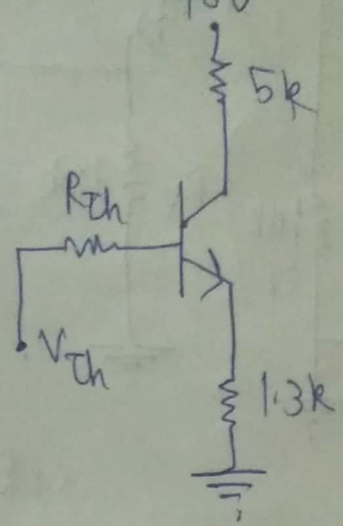
For D C analysis, $\boxed{\omega = 0}$

$$\Rightarrow \boxed{X_C = \frac{1}{\omega C} = \infty}$$

$\therefore$ All Capacitors are open-circuited




eq. $\Longleftrightarrow$




By Thevenin's Theorem —

$$V_{Th} = \frac{\cancel{10}}{\cancel{50}} \times \cancel{10}^{2} = 2V$$

$$R_{Th} = \frac{10 \times 40}{10+40} = 8 k\Omega$$

i/p loop KVL —

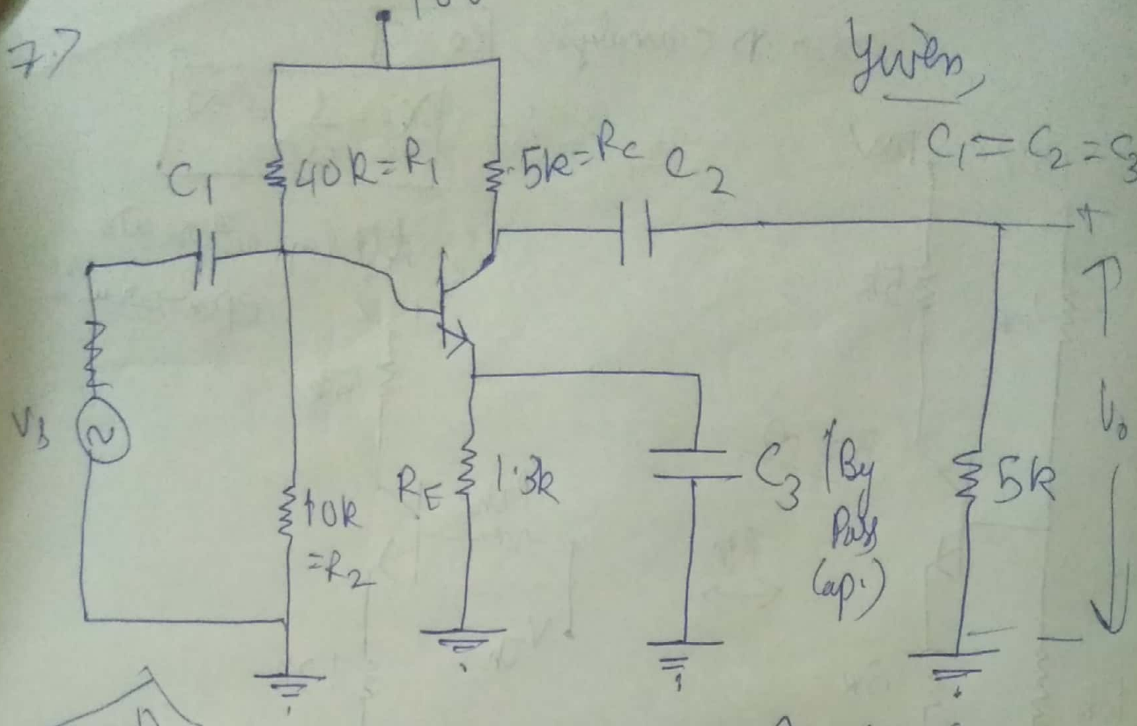
$$V_{Th} - 0 \times R_{Th} - 0.7 - (I_c \times 1.3) = 0$$

$$\left[ \because \beta = \infty \Rightarrow I_B = 0 \Rightarrow \boxed{I_E = I_c} \right]$$

$$I_c = \frac{2 - 0.7}{1.3} \Rightarrow \boxed{I_c = 1mA}$$

$$\Rightarrow \boxed{I_E = 1mA}$$

7.7




Given, $C_1 = C_2 = C_3 = \infty$

$\beta = \infty$

$C_1$ & $C_2$ → Coupling Capacitors

For AC signal, $\omega \neq 0$

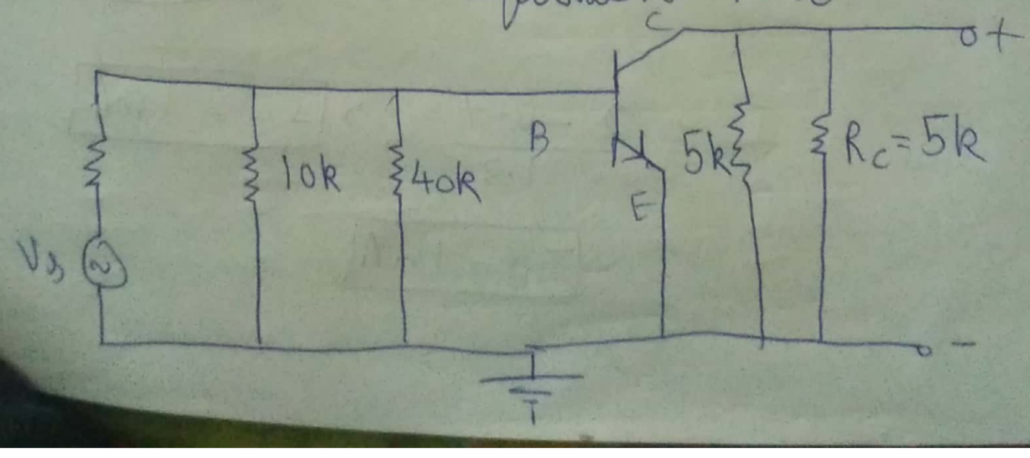
$$X_c = \frac{1}{C\omega} = \frac{1}{\infty} \Rightarrow \boxed{X_c = 0}$$

∴ Capacitors are short-circuited

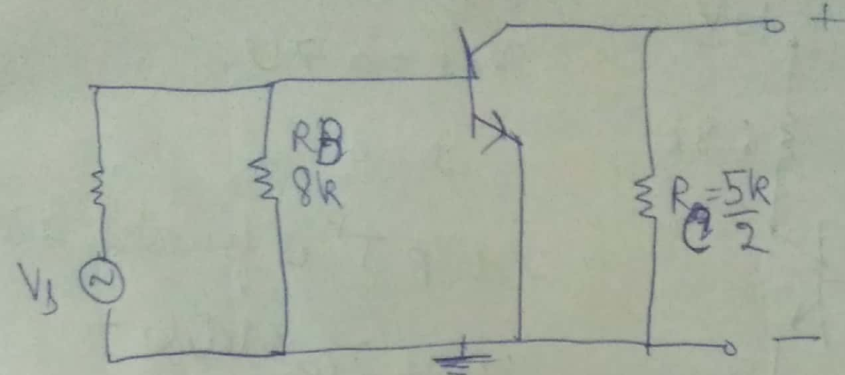
The given circuit is fixed-bias
(for good thermal stability)

We use by-pass capacitor to increase gain
as Gain ↓ when $R_E$↑ (All current pass through $C_3$)

Step I :- Short circuit all dc sources and emitter resistance, as well as all capacitors for ac equivalent model



$R_{eq} = \frac{10\times40}{10+40} = \frac{400}{50} \Rightarrow R_{eq} = 8k$




[given, $V_T = 25$ mV]

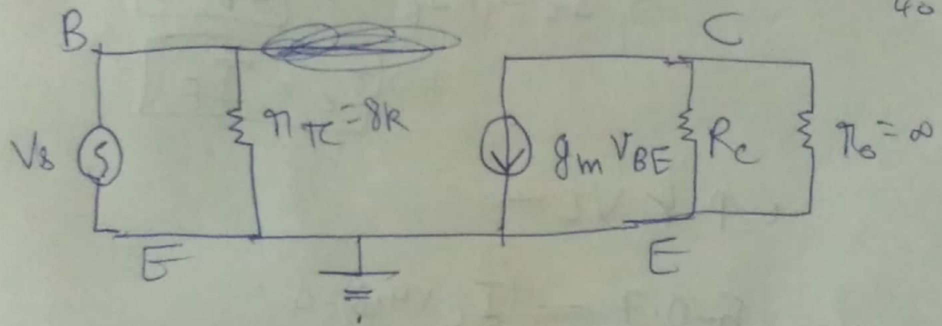
Step: Now, By hybrid-π model (or π model)

$i_c = g_m V_{BE} = i_E$ ($\because \beta = \infty$)

$V_{BE} \approx V_T$ (for small signal)

$\Rightarrow \frac{V_{BE}}{i_E} = \frac{1}{g_m} \Rightarrow g_m = \frac{1}{25mV} = 40\ A/V$

$\Rightarrow r_e = \frac{1}{40}\ \Omega$



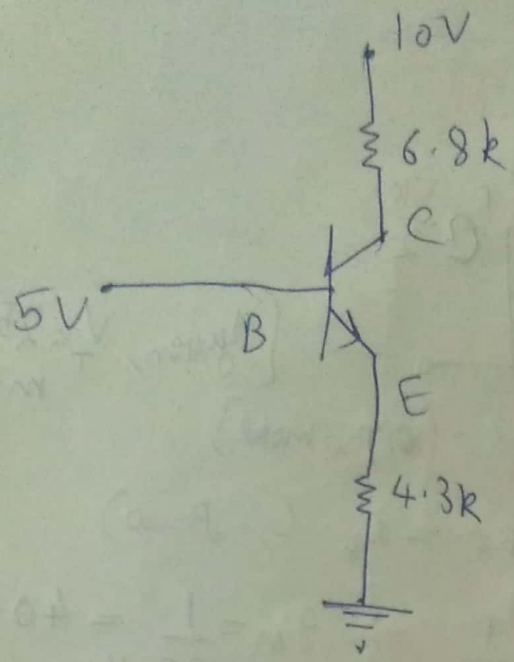

$z_i = R_B \,||\, (\beta+1) r_e = R_B = 8k$

$z_o = r_o \,||\, R_C = \frac{5}{2} \,||\, \infty = \frac{5}{2}k$

$A_V = \frac{V_o}{V_i} = \frac{-\beta \cancel{i_b} R_c}{\cancel{i_b}(\beta+1) r_e} = -\frac{R_c}{r_\pi}$ [$\because \beta \approx (\beta+1)$]

$= -\frac{5}{2\times8}$

$\Rightarrow \boxed{A_V = \frac{-5}{16}}$




Given,

$V_{BE} = 0.7V$

$\beta = \infty$

∵ i/p $J^n$ is forward biased

∴ Only 2 possibilities –

① Sat$^n$ region

② Active region

Assume ② active region –

$\Rightarrow I_C = \beta I_B \Rightarrow \boxed{I_B = 0}$

$\Rightarrow \boxed{I_C = I_E}$

i/p KVL –

$5 - 0.7 - (I_C \times 4.3) = 0$

$\Rightarrow I_C = \frac{4.3}{4.3}$

$\Rightarrow \boxed{I_C = 1mA}$

o/p KVL –

$10 - (6.8 \times 1) - V_{CE} - (4.3 \times 1) = 0$

$\rightarrow \boxed{V_{CE} = -1.1V}$

$V_{CB} = V_{CE} - V_{BE} = -1.1 - 0.7$

$\Rightarrow \boxed{V_{CB} = -1.8V} \Rightarrow V_{CB} < 0$

∴ o/p $J^n$ is forward bias

∴ Our assump$^n$ is wrong

∴ (Sat$^n$ region)

Scanned by CamScanner

9.)

When C & B of a transistor are shorted it acts as a p-n junction diode

Given,

$I_S = 10^{-14} A$

$\eta = 1$

$V_{BE} = 0.7V$

$I = I_S \left( e^{V/\eta V_T} - 1 \right)$

At room temperature,

$V_T = \frac{T}{11600} = \frac{300}{11600} = 0.026V$

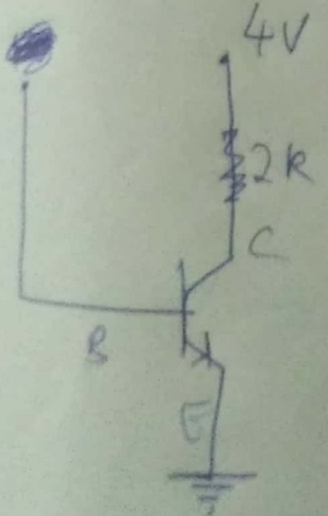
$\therefore I = 10^{-14} \left( e^{(0.7)/0.026} - 1 \right)$

$= 4.9266 \times 10^{-3} A$

$\boxed{I = 4.9266 mA}$

10.)

Given $V_{CE(sat)} = 0.2V$, $\beta_{dc} = 200$




By KVL -

$4 - 2I_C - V_{CE(sat)} = 0$

$\Rightarrow \frac{3.8}{2} = I_C$

$\Rightarrow \boxed{I_C = 1.9mA}$

$\Rightarrow (I_B)_{min} = \frac{I_C}{\beta} = \frac{1.9 mA}{200}$

$\Rightarrow \boxed{(I_B)_{min} = 9.5 \mu A}$